

IILM UNIVERSITY, GREATER NOIDA



INTERNSHIP PRESENTATION

Artificial Intelligence & Machine Learning

STUDENT

Roushan Kant Singh

ROLL NUMBER

25SCS1003002793

PROGRAMME

B.Tech CSE | Batch 2025-29

ORGANIZATION

Codec Technologies Pvt. Ltd.

ROLE

Artificial Intelligence Intern

DURATION

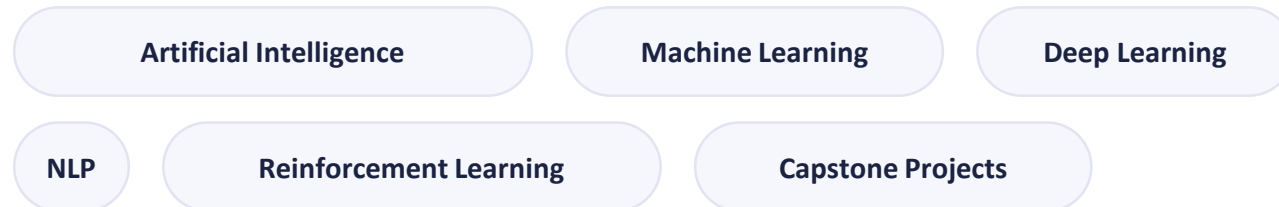
15 Jun 2026 – 15 Jul 2026

INTRODUCTION

Internship Overview

A one-month Artificial Intelligence internship undertaken with Codec Technologies Pvt. Ltd., conducted in association with the National Internship Portal (Ministry of Education) and Google for Education, providing structured, hands-on exposure to the applied AI/ML pipeline.

PROGRAM FOCUS AREAS



ROLE

Artificial Intelligence Intern



DURATION

15/06/2026 – 15/07/2026 (1 Month)



MODE

Pan India, Hybrid

Practical competencies were built across the end-to-end data lifecycle — exploratory analysis, time-series forecasting, CNNs for vision, NLP, Q-learning, and applied capstone engineering.

About Codec Technologies Pvt. Ltd.

A specialized technology software and industrial training enterprise approved by AICTE and ICAC, delivering professional upskilling and applied industrial training across emerging computing paradigms.



AICTE VERIFICATION ID

CORPORATE6759d549ce59e1733940553



NCS IDENTIFICATION

E19E86-0116588288923



QUALITY CERTIFICATION

ISO 9001:2015 Certified



PROGRAM LEADERSHIP

Dr. Anurag Shrivastava (Program Manager / Talent Acquisition Manager)

INSTITUTIONAL PARTNERSHIPS



Google for Education Partner



AICTE & ICAC Approved



**National Internship Portal
(Ministry of Education)**



ISO 9001:2015 Certified Organization

Internship Objectives



Python for AI Development

Programming workflows using NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn



Data Preprocessing & Visualization

Cleaning, transforming, and visualizing multi-variate datasets



Machine Learning Foundations

Supervised learning, model evaluation and optimization



Deep Learning & LSTM Forecasting

Stacked LSTM networks for sequential financial time-series



Natural Language Processing

Text processing and sentiment analysis using NLTK & spaCy



Reinforcement Learning

Q-learning based sequential decision-making



Feature Engineering

Sliding windows, scaling, imputation, SMOTE class balancing



Capstone Implementation

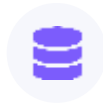
End-to-end applied projects with quantitative model evaluation

Technologies & Tools Used



PROGRAMMING

- Python 3.10+



DATA PROCESSING

- NumPy
- Pandas



VISUALIZATION

- Matplotlib
- Seaborn



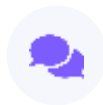
MACHINE LEARNING

- Scikit-Learn
- XGBoost
- Imbalanced-Learn (SMOTE)



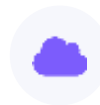
DEEP LEARNING

- TensorFlow
- Keras



NLP

- NLTK
- spaCy



DEV ENVIRONMENT

- Google Colab
- Jupyter Notebook

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TOOLS & LIBRARIES
ACROSS THE PIPELINE

Weekly Curriculum Timeline

WEEK 1	WEEK 2	WEEK 3	WEEK 4
Foundations of AI & Python • AI concepts • Python for AI • Data preprocessing & visualization	Machine Learning Basics • Supervised learning • Model evaluation & optimization	Deep Learning & CNN • Neural networks • CNN fundamentals • Image recognition • Basic CNN model	Advanced AI Applications • NLP, text processing, sentiment analysis (NLTK, spaCy) • Reinforcement Learning, Q-learning • Model evaluation metrics



Capstone Projects

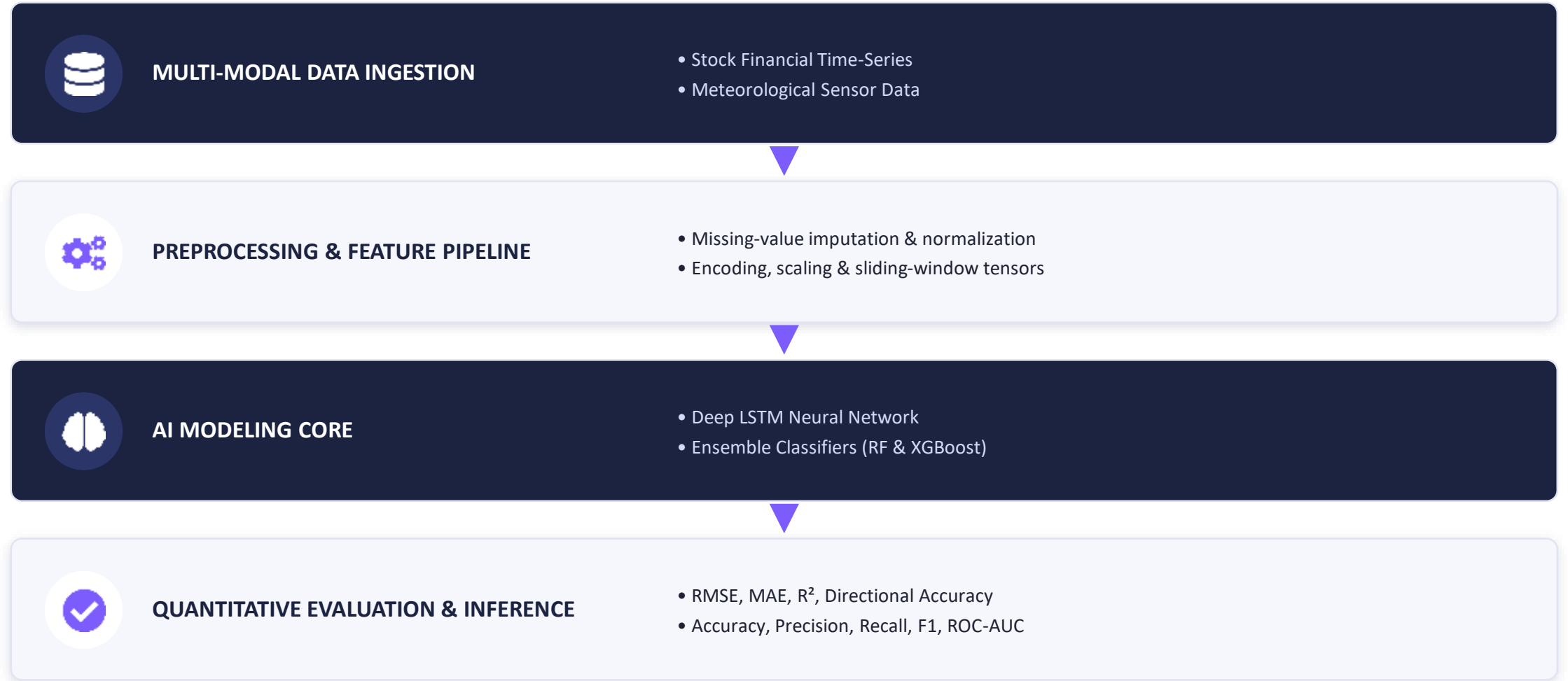


Stock Price Prediction + Weather / Rainfall Prediction



Internship Certification

System Architecture



Stock Price Trend Prediction using Deep LSTM

DOMAIN

FinTech · Deep Learning · Continuous Time-Series Forecasting

PROBLEM

Equity markets exhibit non-linearity, volatility, and long-range temporal dependence that traditional econometric models (ARIMA/GARCH) struggle to capture.

OBJECTIVE

Predict next-day closing price (P_{t+1}) from a historical N -day lookback window using stacked LSTM networks.

DATASET & SPLIT

Open, High, Low, Close, Adjusted Close, Volume · **80% Train / 20% Test (chronological)**

1 Historical Stock Data

2 Preprocessing (MinMaxScaler)

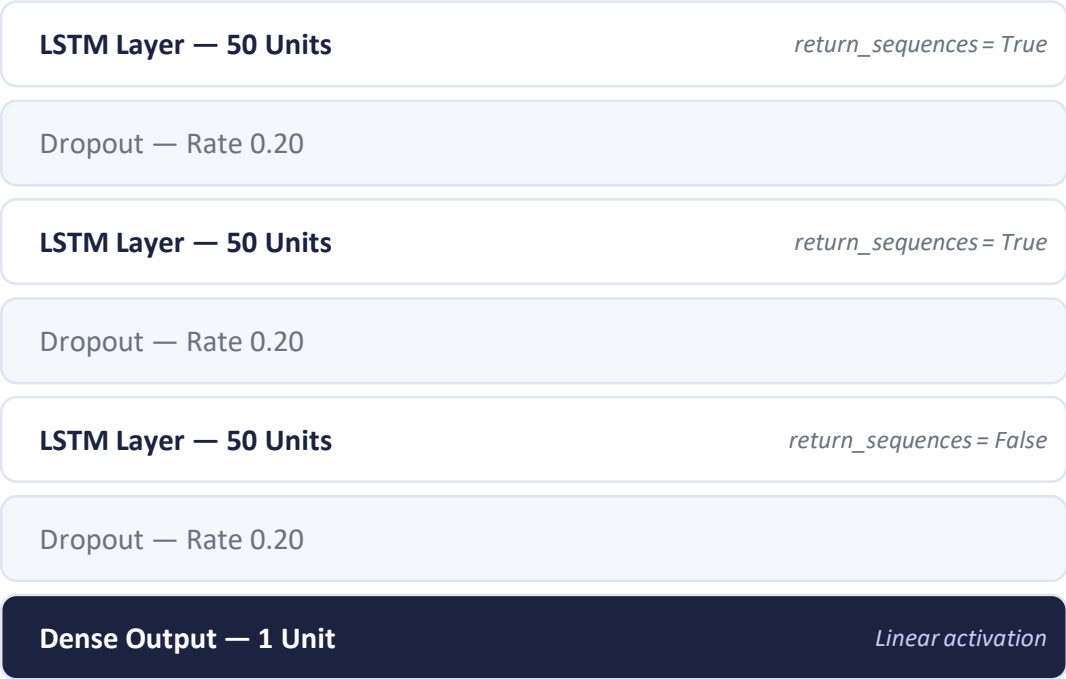
3 60-Day Lookback Window

4 Stacked LSTM Network

5 Price Prediction

6 Evaluation (RMSE, MAE, R^2)

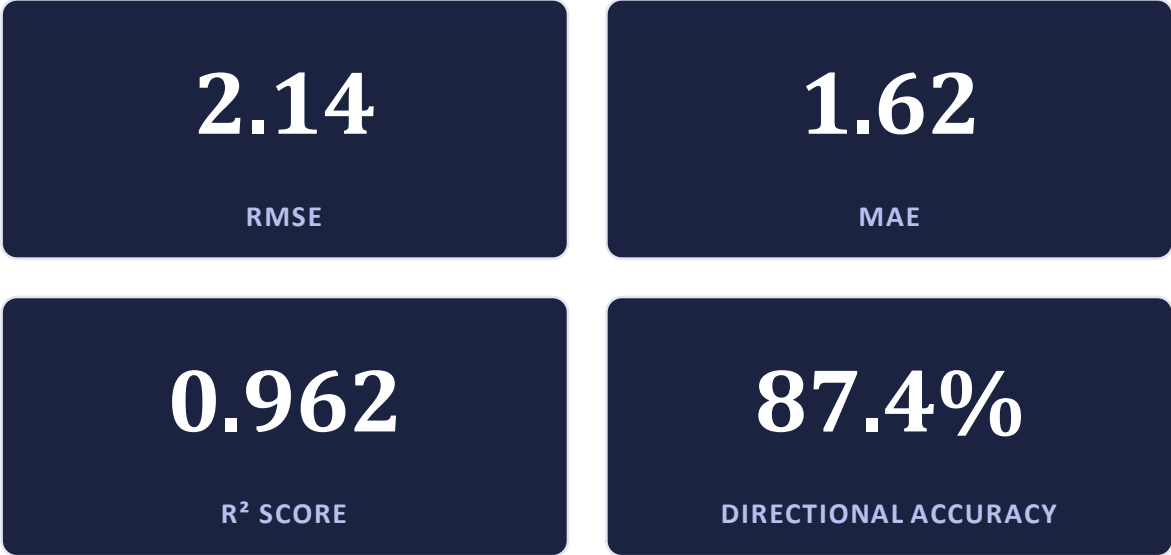
LSTM Architecture & Results



TRAINING CONFIGURATION

50 Epochs · Batch Size 32 · Adam Optimizer · MSE Loss · Early Stopping

TEST-SET PERFORMANCE (OUT-OF-SAMPLE)



The model explains 96.2% of price variance and correctly forecasts next-day direction in 87.4% of test cases.

Weather Data Analysis & Rainfall Prediction

DOMAIN

Environmental Data Science · Applied Meteorology · Supervised Classification

DATASET

140,000+ historical daily weather observations across 23 atmospheric parameters

GOAL

Predict binary precipitation occurrence — Rain Tomorrow: Yes / No

KEY ATMOSPHERIC FEATURES



Temperature



Humidity



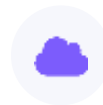
Pressure



Wind Speed



Sunshine



Cloud Cover

PREPROCESSING PIPELINE

- Missing-value imputation (median / modal)
- Outlier capping via IQR method
- Cyclic date & pressure/temperature feature engineering
- One-Hot Encoding of categorical features
- SMOTE class-imbalance correction
- StandardScaler feature scaling

140,000+

DAILY WEATHER OBSERVATIONS USED

Model Benchmark & Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	82.6%	71.3%	52.8%	0.606	0.842
Random Forest (200 Trees)	86.1%	78.4%	61.2%	0.687	0.891
Tuned XGBoost Classifier	88.4%	81.2%	67.5%	0.737	0.924



Best Performing Model: Tuned XGBoost Classifier

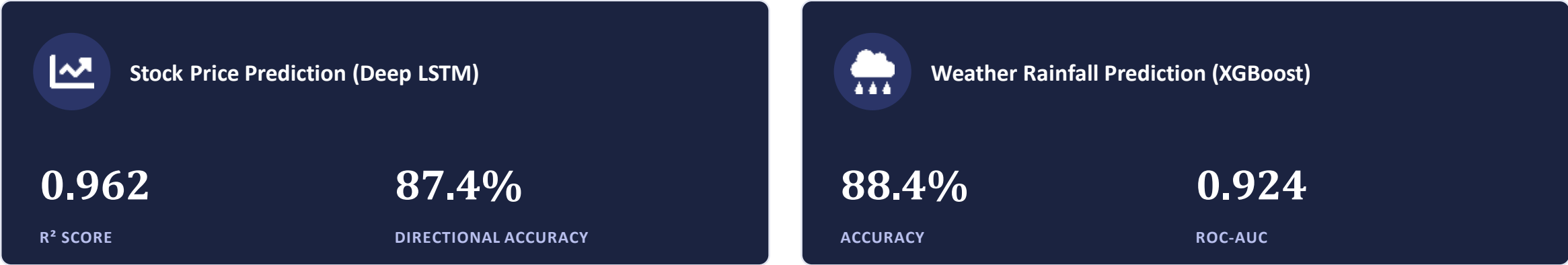
88.4% Accuracy · 81.2% Precision · ROC-AUC 0.924

Key predictive drivers: Humidity (3 PM), Atmospheric Pressure (3 PM), Cloud Cover & Sunshine Duration.

RESULTS SUMMARY

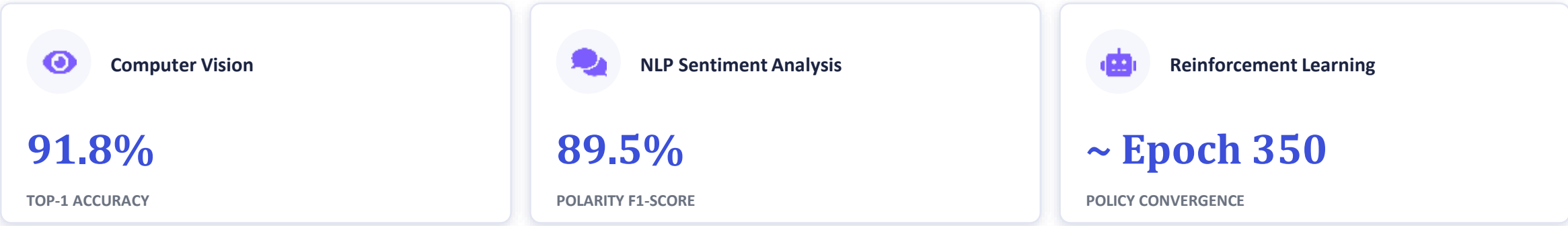
Overall Performance & Achievements

MAJOR CAPSTONE PROJECTS



ADDITIONAL CURRICULUM EVALUATION METRICS

(module exercises within the training curriculum, distinct from the two major capstone projects above)



Weekly Task Completion — Portal Screenshots

Verified screenshots from the Codec Technologies learning portal, confirming progressive completion of all curriculum modules and capstone work.



Figure 4.1: Week 1 — Foundations of AI, Python Development & Data Preprocessing

Week 1 — Foundations of AI & Python



Figure 4.2: Week 2 — Machine Learning Basics, Supervised Learning & Optimization

Week 2 — Machine Learning Basics



Figure 4.3: Week 3 — Introduction to Neural Networks, Deep Learning & CNN Image Models

Week 3 — Neural Networks & CNNs



Week 4 (Part 1) — NLP & Sentiment Analysis



Figure 4.5: Week 4 (Part 2) — Reinforcement Learning (Q-Learning) & Model Evaluation Metrics

Week 4 (Part 2) — Reinforcement Learning



Figure 4.6: Capstone Project Work Completion & Internship Certificate Claim

Capstone Work & Certificate Claim

Internship Certificate & Offer Letter



Internship Completion Certificate

ROLE	ORGANIZATION	DURATION
AI Intern	Codec Technologies Pvt. Ltd.	15/06/2026 – 15/07/2026



Internship Offer Letter

CONCLUSION

Key Takeaways from the Internship



- ✓ Practical, end-to-end understanding of AI/ML workflows
- ✓ Hands-on experience in data preprocessing & model development
- ✓ Deep Learning proficiency through stacked LSTM forecasting
- ✓ Applied ML classification & benchmarking using XGBoost
- ✓ Exposure to CNNs, NLP and Reinforcement Learning
- ✓ Quantitative model evaluation using industry-standard metrics

THANK YOU

Questions & Discussion

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